

Prompt to Anything?

Exploring Generative AI's Iterative Potential in Mapping Show Production

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Figure 1: Regalia Reims AI generated projective texturing. From left to right are textures respectively generated using the normal map, Fresnel map, depth map (see Figure 2) and a combination of those three as input control textures.

ABSTRACT

Incorporating Generative AI (Gen-AI) into our content creation processes at Moment Factory is an adventure driven by curiosity full of excitement and challenges. We have embarked on this journey to not only understand the capabilities of Gen-AI but also to experiment with how to effectively utilize and adapt its creations capabilities to suit our project needs, particularly in the contexts of location-based experiences and 3D mapping shows.

A location-based experience denotes a multimedia interactive and immersive journey meticulously crafted for a specific physical location. As we orchestrate these experiences from inception to execution, we must develop content tailored precisely to each unique setting. These unusual canvases, often vast in scale, are called “mega-canvases”. They can encompass a diverse expansive array of display surfaces integrated into scenography or architectural designs, demanding visual content with hyper-resolutions exceeding 16k.

Our talk aims to unfold our visual and technical explorations, highlighting the innovative integration of Gen-AI into our process, and sharing the significant accomplishments and lessons learned from conceptualization to execution, and the unique challenges

they present. Attendees will leave with a comprehensive understanding of the potential of Gen-AI in enhancing location-based experiences and 3D mapping shows, equipped with knowledge and inspiration to explore these technologies in their own creative endeavors. Through sharing our experiments and outcomes, we aim to foster a dialogue on the future of immersive content creation, inviting others to join us in pushing the boundaries of what is possible with Gen-AI in multimedia experiences.

CCS CONCEPTS

• **Applied computing** → **Media arts; Performing arts; • Computing methodologies** → *Scene understanding; Visual content-based indexing and retrieval; Scene anomaly detection; Image representations; Appearance and texture representations; • General and reference* → *Design*.

KEYWORDS

AI, Art, Computer Vision, Pipeline, Design, Themed Entertainment

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1 GEN-AI AS A CONTENT PRODUCTION TOOL

We use a diffusion model with Stable Diffusion [Rombach et al. 2021] and ControlNet [Zhang et al. 2023] extension in the context of a 3D

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mapping projection project. By inputting a combination of 3D scene related data created from the template model to ControlNet, such as the normal map, Fresnel map and the depth map (Figure 2), we were able to generate scene aware content from template models of Regalia Reims and 25 Broadway (NY, US). We used ComfyUI [Comfyanonymous 2022] for Regalia Reims project and Automatic 1111 [AUTOMATIC1111 2022] for 25 Broadway project as our GUI and backend tool. To maximize iteration time, we generated low scale image and upscaled them using Ultimate SD Upscale [Coyote-A 2023] with ControlNet tile. For example, on 25 Broadway, we generated 3676x512 textures and upscaled them to 15963x2048 textures as shown in Figure 3. The resulting output texture is ready to be mapped to the 25 Broadway model as shown in Figure 4.

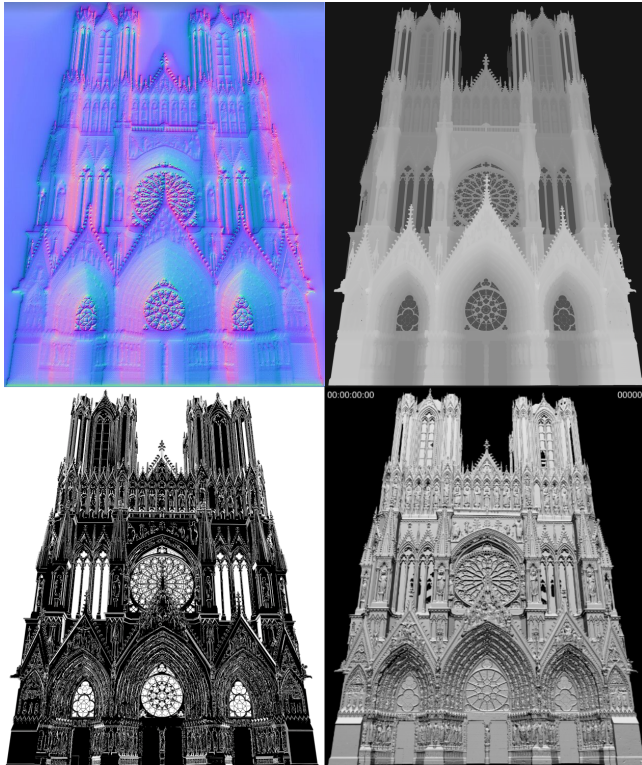


Figure 2: Regalia Reims input images to ControlNet [Zhang et al. 2023]. Top-left is the normal map. Top-right is the depth map defined in world space rather than camera space. Bottom-left is the Fresnel map. Bottom-right is the high fidelity template model used as projection target for AI-generated textures. Since the template model had randomly inverted normals, the normal map was generated from the depth map using Materialize [Software 2018].

2 CONCLUSION

Our experiments have demonstrated the potential of Gen-AI technology to generate images and textures directly from content production templates, enabling rapid iteration of visual concepts and accelerated pre-production pipelines. By leveraging these templates,

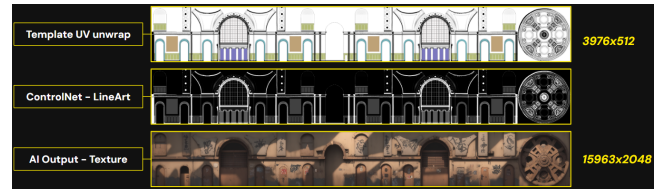


Figure 3: 25 Broadway projection mapping texture generation. From top to bottom texture are respectively : template UV unwrap (3976x512), ControlNet Lineart and the resulting generated AI Output texture upscaled to 15963x2048.



Figure 4: The resulting AI Output texture mapped to 25 Broadway model.

we can quickly explore and refine visual ideas, reducing the time and resources required for concept development. Furthermore, the quality and resolution of the generated content are well-suited for concept and pre-production phases, allowing artists and designers to focus on creative decision-making rather than labor-intensive asset creation. The integration of generated, ready to be mapped, textures into previsualization tools has also shown promise. This has significant implications for the content creation industry, where the ability to quickly iterate and refine ideas can be a key competitive advantage.

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